COST OF CAPITAL

Cost of Equity • Cost of Debt • Weighted Average Cost of Capital

Learning Outcomes

By the end of this session, students will be able to:

2

UNDERSTAND

Bloom's Level 2

Explain the concept of cost of capital and describe the components of cost of equity and cost of debt

3

APPLY

Bloom's Level 3

Calculate cost of equity (CAPM / DGM) and after-tax cost of debt; compute WACC using appropriate weights

4

ANALYZE

Bloom's Level 4

Analyze how capital structure changes affect WACC and evaluate their impact on investment decisions

Features of Equity Shares

1

Residual Claim

Paid last, after debt & preference holders, but owns residual profits/assets

2

Voting Rights

Typically one share = one vote on major corporate decisions

3

Dividends

Not fixed or guaranteed — paid at the board's discretion

4

Perpetual Capital

No maturity date; stays on the balance sheet indefinitely

5

Limited Liability

Losses capped at the amount invested

6

Capital Appreciation

Returns mainly via share price growth, not fixed interest

Ordinary Shares vs. Dual-Class Structures

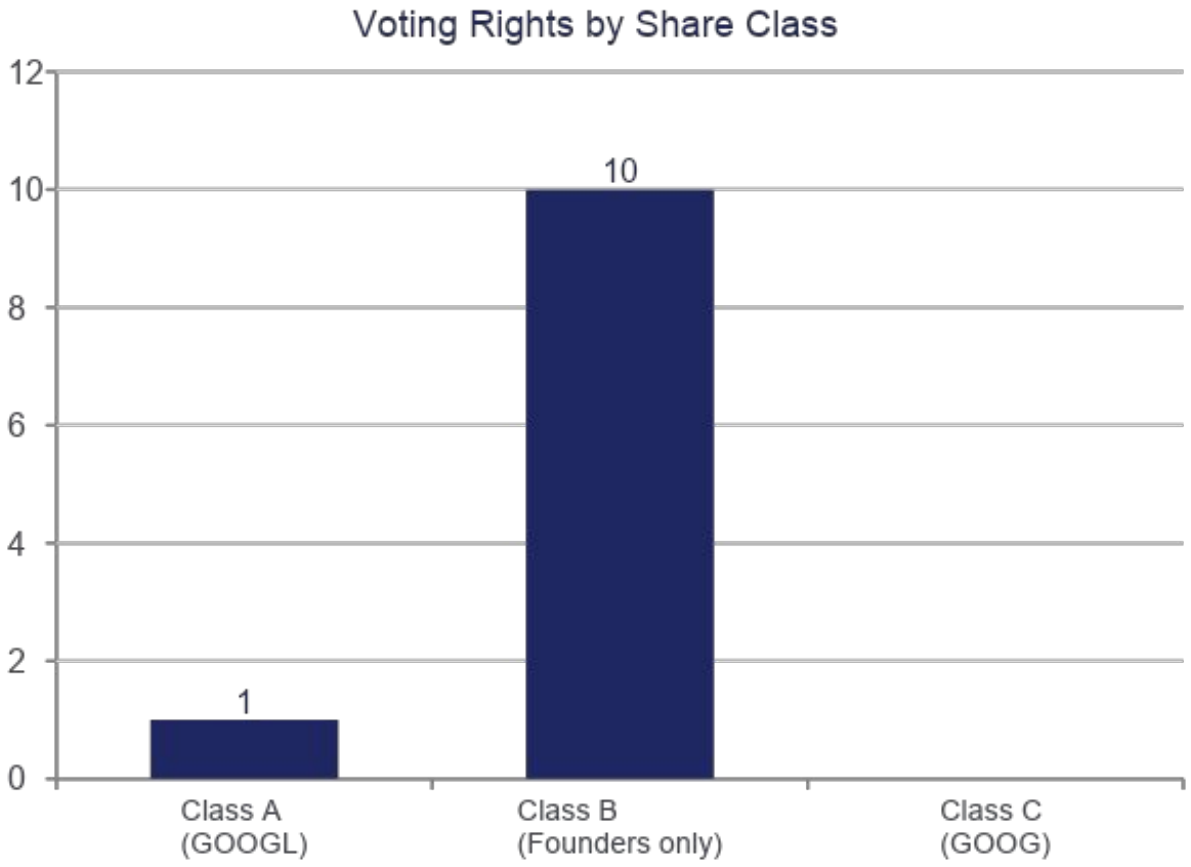
ORDINARY SHARES

- One share = one vote (proportional control)
- Voting power matches economic investment
- Standard structure for most listed companies
- No separate classes with unequal rights

DUAL / MULTI-CLASS SHARES

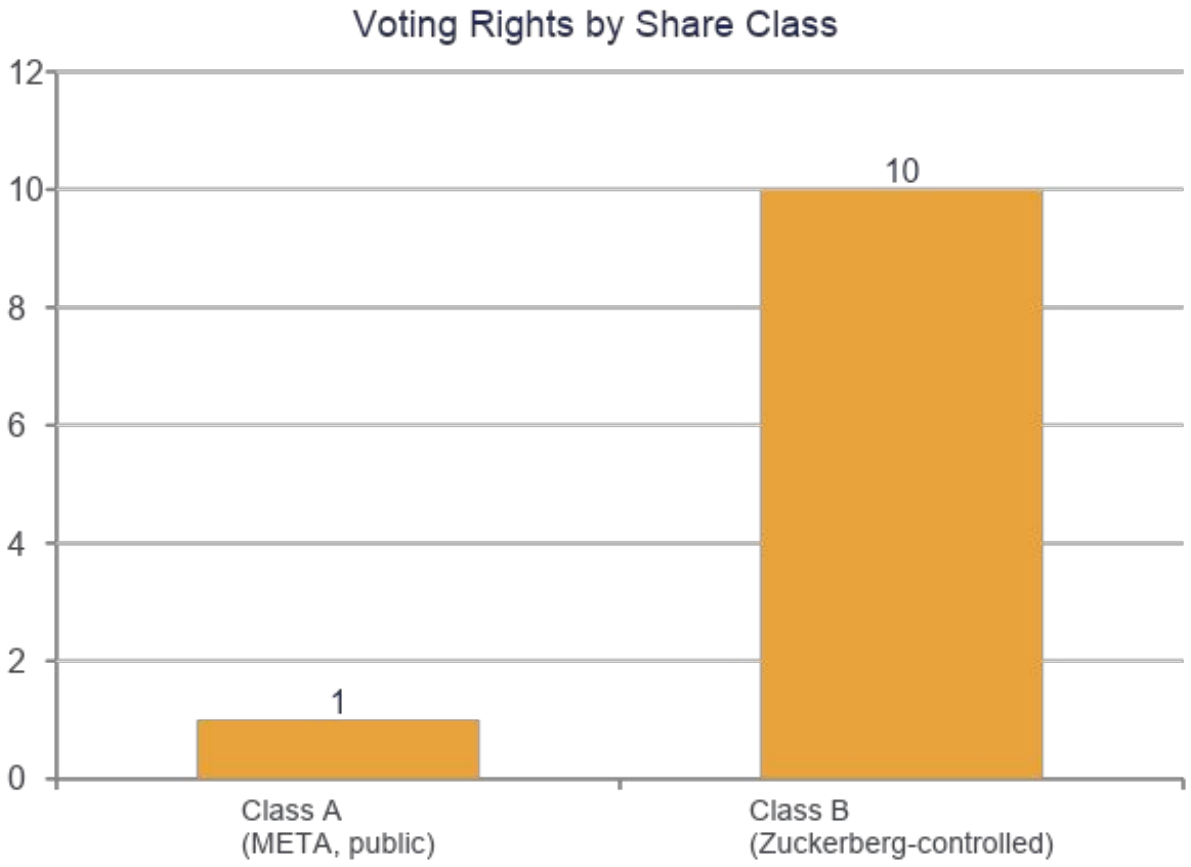
- Same economic (dividend) rights, different voting rights
- Founders hold high-vote shares to retain control
- Public investors typically get low/no-vote shares
- Used by Alphabet, Meta & other tech firms to raise capital without losing control

Case Study: Alphabet's Three-Class Structure



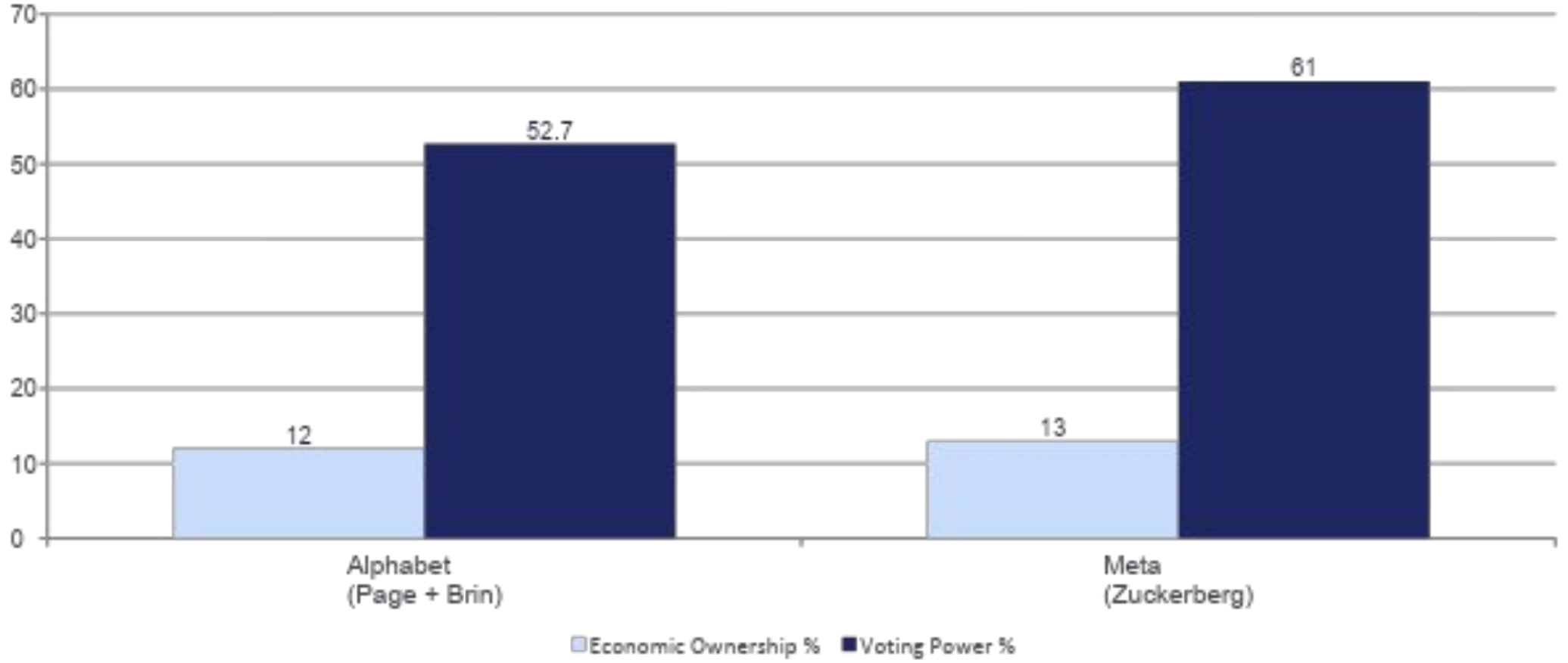
- Class A (GOOGL): 1 vote/share, publicly traded
- Class B: 10 votes/share, held only by founders, never publicly traded
- Class C (GOOG): No votes, publicly traded
- Founders Larry Page & Sergey Brin hold ~52.7% of voting power with under 12% of the economic stake

Case Study: Meta's Two-Class Structure



- Class A (META): 1 vote/share, publicly traded
- Class B: 10 votes/share, held almost entirely by Mark Zuckerberg
- Zuckerberg holds 99.7% of outstanding Class B stock
- This gives him ~61% of total voting power with only ~13-14% of the economic interest

Founder Control: Voting Power vs. Economic Stake



A small economic stake can translate into majority control — the classic separation of ownership and control.

Systematic Risk

Market Risk • Non-Diversifiable Risk

Affects the entire market/economy — arises from macroeconomic, political or social forces no single company controls. Cannot be diversified away.

Interest Rate Hikes

Costlier borrowing depresses valuations market-wide

Inflation

Erodes purchasing power & margins across sectors

Recession (2008 Crisis)

Crashed markets worldwide — banks, retailers, tech alike

Pandemic (COVID-19)

Disrupted supply, demand & labor across every industry

Unsystematic Risk

Specific Risk • Diversifiable Risk

Affects a particular company or industry only — can be reduced or eliminated by holding a diversified portfolio.

Management Decisions

A failed acquisition hurts only that firm's stock

Labor Strikes

Halts one manufacturer's plant, not the whole industry

Product Recall

A defective-drug recall hits one pharma company, not the sector

Fraud / Scandal

An accounting scandal tanks that specific stock

Systematic vs. Unsystematic Risk

Basis	Systematic Risk	Unsystematic Risk
Scope	Entire market / economy	Specific company / industry
Source	Macroeconomic, political, social	Firm-specific (management, labor)
Diversifiable?	No	Yes
Also called	Market risk	Specific / residual risk
Measured by	Beta (β)	Std. deviation of firm-specific returns
Priced by CAPM?	Yes — investors compensated	No — diversified away at no cost

Capital Asset Pricing Model (CAPM)

$$Ke = R_f + \beta \times (R_m - R_f)$$

Ke	**Cost of Equity**	Minimum return equity investors require
R_f	**Risk-Free Rate**	Return on a riskless asset (e.g., 10-yr G-Sec)
β	**Beta**	Sensitivity of the stock to overall market movements
R_m	**Expected Market Return**	Expected return on the broad market index
$(R_m - R_f)$	**Market Risk Premium**	Extra return demanded for bearing market risk

Understanding Beta (β)

β = 1	Moves in line with the market
β > 1	More volatile than market (aggressive: tech, cyclicals)
β < 1	Less volatile than market (defensive: FMCG, utilities)
β = 0	No correlation with market (theoretical, risk-free-like)
β < 0	Moves opposite the market (rare — e.g., gold miners)

CAPM: Examples

Capital Asset Pricing Model (CAPM) & Equity Beta

Estimating Cost of Equity with Real-Life Examples

Understanding CAPM

$$R_e = R_f + \beta\,(R_m - R_f)$$

- **Re** = Cost of Equity
- **Rf** = Risk-Free Rate
- **β** = Equity Beta
- **Rm – Rf** = Market Risk Premium

What is Equity Beta?

$$\beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- Measures sensitivity to market
- **β > 1**: More Volatile than Market
- **β < 1**: Less Volatile (Defensive)



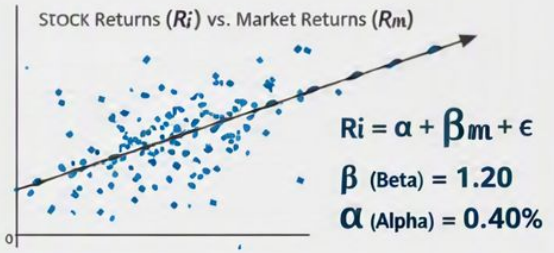
Example: Infosys Cost of Equity

- Risk-Free Rate (R_f) = 7%
- Beta (β) = 1.1
- Market Return (R_m) = 12%

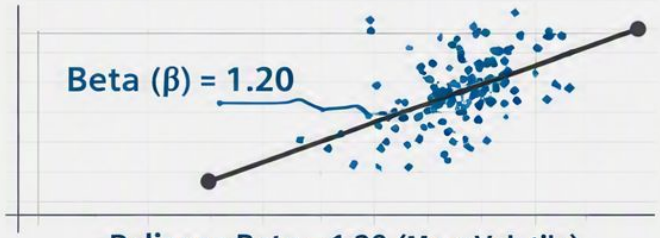
R_f = 7% + 1.1 x (12% - 7%) = **12.5%**

Expected Return = 12.5%

Historical Beta Calculation



Reliance vs Sensex



Reliance Beta = 1.20 (More Volatile)

Conclusion: Key Takeaways

- **CAPM Links Risk & Return**
- **High Beta → Higher Cost of Equity**
- **Low Beta → Lower Cost of Equity**



Historical Equity Beta: Estimation

Covariance / Variance Method

$$\beta = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$$

- $\text{Cov}(R_i, R_m)$: how the stock's returns move with the market's returns
- $\text{Var}(R_m)$: the market's own volatility (variance)

Regression (Slope) Method

$$R_i = \alpha + \beta \times R_m + \varepsilon$$

- α (alpha): stock return unrelated to the market
- β (beta): the slope — systematic risk
- ε (epsilon): error term — firm-specific (unsystematic) risk

Estimating Historical Betas

Item	Selection
Company	Reliance Industries Limited
Stock	Reliance Industries equity shares
Market benchmark	BSE Sensex
Observation frequency	Hypothetical monthly returns
Number of observations	5

Estimating Historical Beta

Month		
January	4%	2%
February	2%	1%
March	−3%	−1%
April	5%	3%
May	−2%	−2%

$$\beta_{\text{Reliance}} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

Historical Equity Beta: Estimation

Step 3: Calculate the Average Stock and Market Returns

Average Reliance return

$$\bar{R}_i = \frac{4\% + 2\% - 3\% + 5\% - 2\%}{5}$$

$$\boxed{\bar{R}_i = 1.2\%}$$

Average Sensex return

$$\bar{R}_m = \frac{2\% + 1\% - 1\% + 3\% - 2\%}{5}$$

$$\boxed{\bar{R}_m = 0.6\%}$$

Historical Equity Beta: Estimation

Step 4: Calculate Deviations from the Average

Month	Reliance return	Sensex return		
January	4%	2%	2.8%	1.4%
February	2%	1%	0.8%	0.4%
March	−3%	−1%	−4.2%	−1.6%
April	5%	3%	3.8%	2.4%
May	−2%	−2%	−3.2%	−2.6%

Historical Equity Beta: Estimation

Step 5: Calculate the Cross-Products and Squared Market Deviations

Month	Stock deviation	Market deviation	Cross-product	Squared market deviation
January	2.8%	1.4%	0.000392	0.000196
February	0.8%	0.4%	0.000032	0.000016
March	−4.2%	−1.6%	0.000672	0.000256
April	3.8%	2.4%	0.000912	0.000576
May	−3.2%	−2.6%	0.000832	0.000676
Total			0.002840	0.001720

Historical Equity Beta: Estimation

Step 6: Calculate Covariance and Variance

For historical sample data:

$$\text{Cov}(R_i, R_m) = \frac{\sum(R_i - \bar{R}_i)(R_m - \bar{R}_m)}{n-1}$$

$$\text{Var}(R_m) = \frac{\sum(R_m - \bar{R}_m)^2}{n-1}$$

Since $n = 5$:

Covariance

$$\text{Cov}(R_i, R_m) = \frac{0.002840}{4}$$

$$= 0.000710$$

Market variance

$$\text{Var}(R_m) = \frac{0.001720}{4}$$

$$= 0.000430$$

Historical Equity Beta: Estimation

Step 7: Calculate Reliance's Historical Beta

$$\beta_{\text{Reliance}} = \frac{0.000710}{0.000430}$$

$$\boxed{\beta_{\text{Reliance}} \approx 1.65}$$

Interpretation

Based on this hypothetical five-month dataset:

- Reliance's estimated beta is approximately 1.65.
- The stock has shown greater sensitivity to market movements than the Sensex.
- A 1% movement in the market would be associated, on average, with an approximately 1.65% movement in the stock in the same direction, based on the estimated relationship.

This is a historical statistical estimate, not a prediction of future returns.

Cost of Equity: Examples

Two BSE Sensex stocks — India 10-yr G-Sec yield ≈ 7.0% used as Rf; assumed Rm = 13%

Reliance Industries (RELIANCE)

β = 0.83

Ke = 7.0% + 0.83 × (13% − 7%)

Ke = 11.98%

Less volatile than the Sensex — modest risk premium

Tata Consultancy Services (TCS)

β = 1.13

Ke = 7.0% + 1.13 × (13% − 7%)

Ke = 13.78%

Slightly more volatile than the Sensex — higher required return

Historical Beta: Example

Illustrative monthly returns: stock vs. BSE Sensex

Month	Stock Return Ri (%)	Sensex Return Rm (%)
1	2.5	1.8
2	-1.2	-0.9
3	3.0	2.2
4	-0.5	-0.3
5	1.8	1.5

Note: Beta varies with the time period, return frequency, and benchmark index chosen — which is why different providers report different beta values for the same stock.

Regression-Based Formula

$$\beta = [n\Sigma(RiRm) - \Sigma Ri \times \Sigma Rm] / [n\Sigma Rm^2 - (\Sigma Rm)^2]$$

1. Find the mean of Ri and Rm
2. Compute deviations from each mean
3. Multiply deviations & sum ⟶ Covariance
4. Square (Rm deviations) & sum ⟶ Variance
5. β = Covariance / Variance

Stock Alpha: Meaning

Alpha measures the **excess return an investment** generates versus what CAPM would predict for its level of risk — it isolates skill from risk-taking.

$$\alpha = R_{actual} - [Rf + \beta(Rm - Rf)]$$

$\alpha = 0$

Performed exactly as expected for its risk

$\alpha > 0$

Outperformed its risk-adjusted expectation

$\alpha < 0$

Underperformed its risk-adjusted expectation

Alpha: Examples

Example 1 — Risk-Adjusted Comparison

Benchmark = 10%, Rf = 4%

	Fund A	Fund B
Beta	1.5	0.8
Actual Return	14%	12%
CAPM Expected	13%	8.8%
Alpha	**+1.0%**	**+3.2%**

Fund B took less risk but delivered far more excess return — more manager skill despite a lower headline return.

Example 2 — Active vs. Passive Fund

Two friends invest ₹1 lakh at similar risk levels

Nifty 50 Index Fund: 20% return

Active Large-Cap Fund: 25% return

Alpha = +5%

That extra 5% reflects the fund manager's stock selection and timing skill — the value added over simply buying the market.

CAPM vs. Dividend Growth Model

Basis	CAPM	Dividend Growth Model
Applicability	All listed companies (dividend or not)	Only dividend-paying companies
Risk measurement	Explicit — via Beta	Implicit / ignored
Data required	Rf, β, Rm	Price, expected dividend, growth rate
Key assumption	Efficient markets	Constant, perpetual dividend growth
Sensitivity	Beta & market premium estimates	Highly sensitive to growth rate (g)
Best suited for	Growth/tech & all-sector firms	Stable, mature, dividend-paying firms

Key Takeaways

- Cost of capital blends cost of equity, cost of debt, and their weights (WACC)
- CAPM prices only systematic (non-diversifiable) risk via Beta
- Alpha reveals genuine skill — excess return beyond what risk alone explains
- CAPM and the Dividend Growth Model are complementary, not competing, tools

Thank You